

Transmission Spectra in EXHALE_transit.py: H α /H β from the Non-LTE $n = 2$ Population, and the Metal Resonance Doublets

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1. Overview

EXHALE_transit.py (the Transmission Probability Model shipped with EXHALE) computes transit transmission spectra by integrating the line-of-sight (LOS) optical depth on an impact-parameter grid, using Voigt profiles evaluated through the Faddeeva function, and then averaging over the projected stellar disk with optional instrument and planet-rotation convolutions. The original code handles the He I 2³S triplet (10830 Å) and the H I / D Ly α doublet (1215.67 Å).

This memo documents the addition of the H α (6562.8 Å, $n = 2 \rightarrow n = 3$) transmission spectrum. Unlike Ly α , which absorbs out of the ground state ($n = 1$) that traces the bulk of the atomic hydrogen, H α absorbs out of the $n = 2$ level,

whose population must be computed from a non-LTE balance. We follow **Christie, Arras & Li (2013, ApJ 772, 144)**, including collisional excitation/de-excitation, recombination cascade, two-photon decay, ℓ -mixing, and **Ly α radiative pumping** ($1s \leftrightarrow 2p$). The Ly α mean intensity $J_{\text{Ly}\alpha}(r)$ is *not* produced by EXHALE; it is either read from an external text file or, by default, estimated with the simple prescription of Huang et al. (2017), see §2.2.

The H α branch always runs; only the source of $J_{\text{Ly}\alpha}$ depends on whether `Jlya_file` is set. The existing He/Ly α behavior is unchanged.

This memo also documents the **metal resonance doublets** (Mg II h&k, Ca II H&K, Na I D; §7.), which are computed by the same LOS pipeline and produce the same three-curve spectrum figures as the H/He lines.

2. The $n = 2$ hydrogen population (Christie et al. 2013)

The $2s$ and $2p$ states are treated separately because they have very different lifetimes (the $2p \rightarrow 1s$ Ly α decay is fast, $A_{2p \rightarrow 1s} = 6.2649 \times 10^8 \text{ s}^{-1}$, whereas the $2s \rightarrow 1s$ two-photon decay is slow, $A_{2s \rightarrow 1s} = 8.26 \text{ s}^{-1}$). With $n_{1s} = n_{\text{HI}}$, n_e and T taken from the EXHALE profiles, the rate-equilibrium equations are (Christie et al. Eqs. 12–13)

$$[A_{2p \rightarrow 1s} + B_{2p \rightarrow 1s} J_{\text{Ly}\alpha} + (C_{2p \rightarrow 1s} + C_{2p \rightarrow 2s}) n_e + \Gamma_{2p}] n_{2p} - C_{2s \rightarrow 2p} n_e n_{2s} = (B_{1s \rightarrow 2p} J_{\text{Ly}\alpha} + C_{1s \rightarrow 2p} n_e) n_{1s} + \alpha_{2p} n_e^2, \quad (1)$$

$$[(C_{2s \rightarrow 1s} + C_{2s \rightarrow 2p}) n_e + \Gamma_{2s} + A_{2s \rightarrow 1s}] n_{2s} - C_{2p \rightarrow 2s} n_e n_{2p} = C_{1s \rightarrow 2s} n_e n_{1s} + \alpha_{2s} n_e^2. \quad (2)$$

This is a 2×2 linear system in (n_{2p}, n_{2s}) ,

$$\begin{pmatrix} L_{2p} & -C_{2s \rightarrow 2p} n_e \\ -C_{2p \rightarrow 2s} n_e & L_{2s} \end{pmatrix} \begin{pmatrix} n_{2p} \\ n_{2s} \end{pmatrix} = \begin{pmatrix} S_{2p} \\ S_{2s} \end{pmatrix},$$

solved per radial cell by Cramer's rule. The total $n = 2$ density driving H α absorption is $n_2 = n_{2s} + n_{2p}$; in practice it is dominated by the metastable $2s$ state.

2.1 Rate coefficients

The forward collisional rates are taken from Christie et al. Table 2 (Janev et al. 2003); the reverse (super-elastic) rates follow by detailed balance using the statistical weights $g_{1s} = 2$, $g_{2s} = 2$, $g_{2p} = 6$. With $t_4 \equiv T/10^4 \text{ K}$:

Process	Coefficient	Expression [$\text{cm}^3 \text{ s}^{-1}$]
Case-B recomb. (Draine 2011)	α_B	$2.54 \times 10^{-13} t_4^{-0.8163-0.0208 \ln t_4}$
$\rightarrow 2s$ (incl. cascade)	α_{2s}	$(0.282 + 0.047 t_4 - 0.006 t_4^2) \alpha_B$
$\rightarrow 2p$	α_{2p}	$\alpha_B - \alpha_{2s}$
$1s \rightarrow 2s$ excitation	$C_{1s \rightarrow 2s}$	$1.21 \times 10^{-8} t_4^{-0.455} e^{-118400/T}$
$1s \rightarrow 2p$ excitation	$C_{1s \rightarrow 2p}$	$1.71 \times 10^{-8} t_4^{-0.077} e^{-118400/T}$
$2s \leftrightarrow 2p$ ℓ -mix	$C_{2s \rightarrow 2p}$	$6.21 \times 10^{-5} (\ln(T/T_0) - \gamma) / \sqrt{T}$
$2s \rightarrow 1s$ (super-elastic)	$C_{2s \rightarrow 1s}$	$1.21 \times 10^{-8} t_4^{-0.455} (g_{1s}/g_{2s})$
$2p \rightarrow 1s$ (super-elastic)	$C_{2p \rightarrow 1s}$	$1.71 \times 10^{-8} t_4^{-0.077} (g_{1s}/g_{2p})$
$2p \rightarrow 2s$ ℓ -mix	$C_{2p \rightarrow 2s}$	$C_{2s \rightarrow 2p} (g_{2s}/g_{2p})$

with $T_0 = 1.02\text{ K}$ and $\gamma = 0.57721$ (Euler–Mascheroni). The reverse rates are written in their analytically-cancelled form: the Boltzmann factor $e^{+118400/T}$ implied by detailed balance cancels the $e^{-118400/T}$ inside $C_{1s \rightarrow 2s}/C_{1s \rightarrow 2p}$, which avoids a numerical $0 \times \infty$ at the very low temperatures that can occur at the base of the post-processed profile.

2.2 Ly α pumping and the $J_{\text{Ly}\alpha}$ input

The $1s \rightarrow 2p$ pumping rate is $B_{1s \rightarrow 2p} J_{\text{Ly}\alpha}$, with the Einstein coefficient in the mean-intensity convention

$$B_{1s \rightarrow 2p} = \frac{g_{2p}}{g_{1s}} \frac{c^2}{2h\nu_{\text{Ly}\alpha}^3} A_{2p \rightarrow 1s} \simeq 8.55 \times 10^9 \text{ cgs},$$

and the stimulated-emission rate $B_{2p \rightarrow 1s} J_{\text{Ly}\alpha}$ with $B_{2p \rightarrow 1s} = (g_{1s}/g_{2p})B_{1s \rightarrow 2p}$. Here $J_{\text{Ly}\alpha}$ is the **angle-averaged mean intensity J_ν at Ly α line center**, in cgs units $\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}$, so that $B_{1s \rightarrow 2p} J_{\text{Ly}\alpha}$ has units of s^{-1} .

$J_{\text{Ly}\alpha}(r)$ is obtained in one of two ways.

(1) From a file. If `Jlya_file` points to an existing two-column text file, $J_{\text{Ly}\alpha}(r)$ is read from it:

column 1	r/R_p	(same radial coordinate as the EXHALE output)
column 2	$J_{\text{Ly}\alpha}$	$[\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}]$

linearly interpolated onto the EXHALE grid and clamped to the endpoints. A template ships as `inputdata/Jlya.txt.example`.

(2) Simple estimate (default; Huang et al. 2017). If no file is given, $J_{\text{Ly}\alpha}$ is estimated following Huang, Arras, Christie & Li (2017, ApJ 851, 150), Eq. (6) and the surrounding text,

$$J_{\text{Ly}\alpha}(r) \simeq 0.1 \frac{F_{\text{LyC}}}{\Delta\nu_D(r)}, \quad \Delta\nu_D = \nu_{\text{Ly}\alpha} \frac{1}{c} \sqrt{\frac{2k_B T}{m_p}},$$

where $\Delta\nu_D$ is the local Ly α Doppler width and F_{LyC} is the *deposited* (absorbed) Lyman-continuum energy flux. The prefactor encodes the assumption that each LyC photoionization is balanced by a recombination producing one Ly α photon; Huang et al. identify F_{LyC} with the total LyC flux absorbed inside the atmosphere ($2.6 \times 10^4 \text{ erg cm}^{-2} \text{ s}^{-1}$ for HD 189733b, page 11).

F_{LyC} from the actual stellar input. Rather than a fixed number, EXHALE_transit.py computes the absorbed LyC flux from the EXHALE inputs:

$$F_{\text{LyC}} = \xi \underbrace{\frac{10^{L_{\text{EUV}}}}{4\pi a^2}}_{\text{incident stellar LyC}} \times \underbrace{(1 - e^{-\tau_{\text{LyC}}})}_{\text{fraction absorbed}}, \quad \tau_{\text{LyC}} = \sigma_{\text{LyC}} N_{\text{HI}},$$

where L_{EUV} (the EUV-band luminosity, $E > 13.6\text{ eV}$) and the orbital distance a are read from `input.inp`, N_{HI} is the vertical neutral-hydrogen column from the EXHALE profile, and $\sigma_{\text{LyC}} \simeq 6.3 \times 10^{-18} \text{ cm}^2$. The factor ξ is the day-night / 2D flux dilution, taken to match ATEs’s “2D approximate method” (`Rate/2` $\rightarrow$ 0.5, `Rate/4` $\rightarrow$ 0.25, else 1; read from `input.inp`, cf. the ξ factor of Christie+2013 / Huang+2017). For the optically-thick atomic layer $\tau_{\text{LyC}} \gg 1$, so the absorbed fraction $\rightarrow 1$ and $F_{\text{LyC}} \rightarrow \xi \times$ the incident stellar LyC flux (e.g. $\xi = 0.5$, $\simeq 5.2 \times 10^2 \text{ erg cm}^{-2} \text{ s}^{-1}$ for the HD 209458b input). A fixed value can be forced via `F_LyC_override` and the dilution via `xi_override`.

This estimate is a uniform-illumination approximation: it omits the decline of $J_{\text{Ly}\alpha}$ deep in the atmosphere captured by the full radiative-transfer treatment (Huang et al. Eq. 7) and the direct stellar Ly α component; for a self-consistent profile, supply Jlya_file.

2.3 $n = 2$ photoionization Γ_{2s}, Γ_{2p} (Balmer continuum)

The $n = 2$ levels are photoionized by the stellar Balmer continuum ($E > 3.4$ eV, $\lambda < 3647$ Å); this depletes n_2 and is included in Eqs. (12)–(13) through Γ_{2s}, Γ_{2p} . The Balmer-continuum field is outside EXHALE's XUV scope, so it is estimated from a diluted stellar blackbody: for a stellar effective temperature T_{star} and radius R_{star} ,

$$\Gamma_2 = \int_{\nu_2}^{\infty} \frac{F_{\nu}}{h\nu} \sigma_2(\nu) d\nu, \quad F_{\nu} = \pi B_{\nu}(T_{\star}) \left(\frac{R_{\star}}{a}\right)^2, \quad \sigma_2(\nu) = \sigma_{2,\text{th}} \left(\frac{\nu_2}{\nu}\right)^3,$$

with ν_2 the $n = 2$ threshold and $\sigma_{2,\text{th}} \simeq 1.4 \times 10^{-17} \text{ cm}^2$, and $\Gamma_{2s} \simeq \Gamma_{2p} \simeq \Gamma_2$. This reproduces Huang et al.'s Table 2 values (~ 20 – 26 s^{-1}) for a K dwarf, and is substantially larger for a hotter G dwarf (e.g. $\sim 140 \text{ s}^{-1}$ for HD 209458's $T_{\star} \simeq 6065 \text{ K}$), which suppresses n_2 and hence the Balmer-line depth. Setting $T_{\text{star}} \leq 0$ falls back to the manual constants Gamma_2s/Gamma_2p (default 0).

3. H α and H β optical depth and transmission

Once $n_2(r)$ is known, the H α transmission is computed with the same LOS integration as Ly α . Along a chord at impact parameter b ,

$$\tau_{\nu}(b) = \int n_2(r) \sigma_{\text{H}\alpha}(\nu, r) d\ell, \quad \sigma_{\text{H}\alpha} = \frac{\sqrt{\pi} e^2}{4\pi\epsilon_0 m_e c} \frac{f_{23}}{\Delta\nu_D} H(a, x),$$

where $H(a, x) = \text{Re}[w(x + ia)]$ is the Voigt function (Faddeeva w), $\Delta\nu_D = \nu_{\text{H}\alpha} v_{\text{th}}/c$ uses the proton thermal speed, and the line-of-sight bulk velocity is folded into x exactly as for the existing lines. Atomic data: $\lambda_{\text{H}\alpha} = 6562.8$ Å (air), $f_{23} = 0.6407$, $A_{3 \rightarrow 2} = 4.41 \times 10^7 \text{ s}^{-1}$ (the line is strongly Doppler-dominated, so the damping wing is negligible). The disk-averaged transmission, instrument convolution and planet-rotation convolution reuse the existing code paths.

H β ($n = 2 \rightarrow n = 4$). The same $n = 2$ population also produces H β at $\lambda_{\text{H}\beta} = 4861.35$ Å ($f_{24} = 0.119$, $A_{4 \rightarrow 2} = 8.42 \times 10^6 \text{ s}^{-1}$). It is computed with an identical LOS loop and plotted/printed alongside H α . Because both lines are optically thick (saturated), the H α /H β depth ratio is much smaller than the optical-depth ratio (~ 7.3): e.g. on the converged HD 209458b run, H $\alpha \simeq 3.2\%$ and H $\beta \simeq 2.0\%$ with TPM's default star, and $\simeq 0.41\% / 0.26\%$ with HD 209458's actual $R_{\star}, T_{\star}$ (see the comparison notebook).

4. Implementation in EXHALE_transit.py

System parameters. The planet and the star are taken from the run directory, matched by label exactly as input_read.f90 does. Besides R_p , M_p , T_{eq} , a , $M_{\star}$, L_{EUV} and the 2-D approximate method, this includes the stellar radius (Stellar radius [R_sun]) and effective temperature (Stellar Teff [K]), which set the transit normalization $A_{\star}$, the $R_{\star}/R_p$ cap on the absorbing annulus, and the diluted-blackbody Balmer continuum of §2.3. The planet spin period has no input.inp key: a close-in giant is tidally locked, so it defaults to the Keplerian orbital

period $P_{\text{orb}} = 2\pi \sqrt{a^3 / [G(M_* + M_p)]}$. Each of the three is resolved as EXHALE_TRANSIT_RSTAR_RSUN / _TSTAR / _ROTP environment override > ./input.inp > built-in default, and the resolved values and their sources are printed at startup.

EXHALE_resolved.out comes first (2026-08-22). $R_p, M_p, T_{\text{eq}}, a, M_*$ and He/H are read from EXHALE_resolved.out beside input.inp whenever that file exists, and only from input.inp when it does not. The wind solver writes it (write_resolved_config in write_setup_report.f90) *after* a base.inp handoff or a lower-atmosphere profile has moved the base, so it is the geometry the wind actually used, while input.inp alone is stale in exactly that case. read_input_params says at startup which of the two it took, and a file that is present but unreadable raises rather than falling back silently. The stellar radius and T_{eff} are not in that file and keep the precedence above.

- $n2_populations(T, n1s, n\text{HII}, ne, J_{\text{Ly}\alpha}, G2s, G2p)$ solves the 2×2 system above and returns (n_{2s}, n_{2p}, n_2) in cm^{-3} . The Ly α -pump constants $B_{1s \rightarrow 2p}, B_{2p \rightarrow 1s}$ are precomputed.
- $J_{\text{Ly}\alpha}$ is read from Jlya_file if present, otherwise built from the Huang et al. (2017) estimate (§2.2); n_2 is then built into the symmetric (inverted+normal) radial profile data_n2 (in m^{-3}), exactly like data_nHI.
- A new wavelength grid l_onda_Ha and an H α LOS loop fill exp_tau_Ha; the disk average, instrument convolution (Instr_res_Ha) and rotation convolution produce the plotted curves.
- The H α branch always runs (do_Ha=True); with the default Jlya_file='' it uses the Huang et al. (2017) estimate for $J_{\text{Ly}\alpha}$, with F_{LyC} taken as the absorbed stellar LyC flux (§2.2).
- **Compatibility fix.** Ion_species.txt in EXHALE carries trace-metal columns, so the reader was changed to usecols=range(7) to take only r and the six H/He species. This also remains correct for the old 13-column (C/O-only) and 7-column (H/He-only) outputs. The column count has since grown: write_output.f90 writes r , the six H/He species (HI HII HeI HeII HeIII HeITR) and all n_mion = 27 metal-ion stages, i.e. **34** columns, and **38** on a molecular run, which appends H2 H2p H3p HeHp; the metal columns are written whether or not metals are on. The usecols=range(7) slice is unaffected (the first seven columns are still r plus the six H/He species), and anything past them is found through the file's own # columns schema header.

4.1 Outputs

Every line carries one key (He10830, Lya, Halpha, Hbeta, MgII, CaII, NaI), and both products of a line are named from that key and land in the run directory:

Product	Name	Default
model curve	<path>/tpm_<line>.txt	always written
figure	<path>/<prefix><line>.png (+ .pdf)	off; set EXHALE_TRANSIT_FIG_PREFIX

A curve file has the columns $\lambda[\text{\AA}]$, T_λ (theoretical), T_λ (instrument-convolved), T_λ (+rotation), so the excess absorption in percent is $(1 - T_\lambda) \times 100$. The set follows the run: a metals-off run writes no metal curves. EXHALE_TRANSIT_SAVE_PREFIX decorates the curve name exactly as the figure prefix decorates the figure name, and

both are resolved against the run directory, so an absolute prefix writes elsewhere and nothing depends on the working directory. Every curve file also opens with a block of # comment lines stating what the spectrum was built on: the input profile and its validity schema, the census of the rows the chords sampled, and the overrides in effect (§5.3). They are comments, so the four numerical columns are what they always were.

5. What the spectrum is built on: row validity, the census, and the metadata block

The spectra are integrated over the `*_adv.txt` profiles, and those profiles do not carry one uniform product: the steady advective correction is applied to some rows and refused on others. Since the profiles began stating that row by row, the transit tool reads the statement, reports what the chords sampled, and writes both into every curve it saves. None of this changes a numerical column of a `tpm_*.txt` file.

5.1 Reading the profile's own statement of validity

`read_adv_validity(exhale_transit_lib.py)` decides which of three cases a profile is, by its header and by nothing else. Columns are located by name on the # columns line, so a file that gains or reorders columns is still read correctly.

adv_schema	What the file carries	Row decision
2	adv_T_status, adv_comp_status, adv_mass_row	refused when adv_T_status $\neq 0$
1	one adv_status column	refused when it is nonzero
absent	no schema line and no status column	every row UNKNOWN

Under schema 2 each status field takes one of five values, 0 corrected, 1 retained, 2 failed, 3 unsupported, 4 not_evaluated. The temperature field is what the census refuses on; the composition field is reported beside it, because a corrected temperature can sit on a retained composition and the lower-level density of every line comes from that composition.

A corrected row is a *conditional* correction, not a certified one. The profile states the fraction it is accurate to in the mass flux, `adv_conditional_tol` (currently 10^{-2} , the first-order argument at its declaration), and the measure each row was decided by, `adv_mass_row` = $|R_1|/s_1$, the mass row of the certification's own residual scaled by that row's own largest term. The header's `adv_input_certified` line is a different statement and is kept separate: it is the certification's verdict on the run state, at the much tighter `cert_tol_mass`, not a verdict on the post-processed rows. The reader returns the tolerance, its header text and the measure column, so the spectrum can carry the condition of the rows it stands on.

A legacy profile is read as UNKNOWN and said to be UNKNOWN. It is not read as "corrected": a writer that did not state its refusals says nothing about them, and a census of such a file can only say so. Both the console and the saved metadata print that distinction explicitly, because UNKNOWN is not zero refused.

5.2 The census: which rows the chords sample

A ray at impact parameter b crosses every shell with $r \geq b$, at line-of-sight coordinate $|x| = \sqrt{r^2 - b^2}$, and there is *no upper radial limit*: a shell at $r = 12$ meets the ray at $b = 2$ whether or not the ray grid stops at $b = 10$, so a shell lying outside the stellar disk still absorbs on the rays that cross the disk. One function, `chord_shell_indices`, now serves the census, `resonance_depth` and `resonance_spectrum` alike; two selections would let the diagnostic describe a different atmosphere from the one the spectrum was built from. The earlier census mask imposed an upper limit at the stellar-disk cap R_{ib} and therefore described less material than the spectrum contains. That defect is latent whenever R_{ib} is the outermost grid radius, which both regression profiles are, and real as soon as the domain passes the stellar radius: MEASURED on `wasp_full` read at `EXHALE_TRANSIT_RSTAR_RSUN=0.30`, the sampled rows go from 419 to 500, 81 of them above the cap.

The console reports, and the metadata block records, how many of the sampled rows are not the steady advective correction, broken down by reason and with the composition field beside it. Then, line by line at eight impact parameters and at its maximum, it reports the share of the line-center optical depth carried by those rows. The share is exact rather than estimated: the optical depth of a ray is the trapezoid sum $\tau(b) = \sum_k w_k I_k$ over the rows the chord crosses, which is additive, so a subset of the rows carries exactly $\sum_{k \in \text{subset}} w_k I_k / \tau(b)$ (`refused_line_center_tau_share`).

It is a CONTRIBUTION, not an uncertainty. The disk-averaged transmission is an average of $\exp(-\tau)$ and is nonlinear in τ , so a saturated ray moves very little when its optical depth moves, and a share near one does not mean the depth is wrong by a comparable amount. The tool labels it as such in the console and in the file, and it should be quoted that way. On `wasp_full` the rays near $b = 1$ take 99 percent of their Ly α line-center depth from uncorrected rows (MEASURED), which says where the material is, not that the depth is unreliable.

5.3 The metadata block on every saved curve

`transit_metadata_block` writes a block of # comment lines at the head of every `tpm_*.txt` product, the two handwritten products included. Comments only: `np.loadtxt` and any reader that skips comment lines sees exactly the numerical columns it saw before, and the numerical columns were verified bitwise across all ten `wasp_full` and five `mol_base_handoff` products when the block was introduced. The block carries, in order:

- `transit_schema` (currently 1) and a one-line statement that the product is a one-way transmission of the profile named next;
- the input profile's path, its `adv_schema`, its `adv_input_certified` verdict, its stationarity operator, its `adv_conditional_tol` text and the largest `adv_mass_row` over its rows, and its model restrictions, all copied out of the profile's own header, so a reader of the curve alone can tell what it stands on;
- the profile's provenance lines;
- the tool's own identity, by git hash and file modification times;

- the census for this line: sampled rows, refused rows, rows above the disk cap, the refusal reasons, the composition breakdown, the eight-point refused-share ladder and its maximum with the b it occurs at, and the sentence saying the share is a contribution and not an uncertainty;
- every EXHALE_TRANSIT_* or TPM_* override in effect, or a line saying that none is and every window, resolving power and geometry setting is the built-in default or input.inp.

When the input states no row validity the census line says so, in the words “the count is UNKNOWN, which is not zero refused”.

6. Validation

Run end-to-end on the converged HD 209458b EXHALE output (output/*_adv.txt), the module is stable and NaN-free and yields clean H α and H β absorption lines at 6562.8 and 4861.4 Å. With the Huang $J_{\text{Ly}\alpha}$ estimate ($\xi = 0.5$, $F_{\text{Ly}\alpha} \simeq 5.2 \times 10^2$, Γ_2 from $T_* = 6065$ K), the depths are physically reasonable and the H α /H β ratio reflects saturation (both optically thick). The Ly α branch in the same run saturates ($\sim 100\%$ core absorption) and He I behaves as before. A worked comparison against the Jensen et al. (2012) HD 209458b observation is provided in the notebook Halpha_compare_HD209458b.ipynb.

7. Metal resonance doublets (Mg II h&k, Ca II H&K, Na I D)

The metal resonance lines are the key NUV/optical diagnostics of metal-enriched escaping atmospheres (Huang et al. 2023). They absorb out of the *ion ground state*, whose excited fine-structure levels are Boltzmann-negligible at $\sim 10^4$ K, so the lower-level density is the ion density itself ($n_{\text{lower}} \simeq n_{\text{ion}}$): no non-LTE population calculation is needed. The ion-density profiles are read from the metal block of Ion_species_adv.txt; a metals-off run (all-zero columns) skips this section automatically.

7.1 Atomic data and implementation

Each species is treated as a **doublet**, with both components summed in a single wavelength window (Table 1; NIST f -values and A -coefficients). The optical depth uses exactly the H/He pipeline: spherical chords $\sqrt{r^2 - b^2}$ on the impact-parameter grid, LOS velocity $v_{\text{los}} = vx/r$, Voigt profiles via the Faddeeva function (vectorized over wavelength), trapezoidal τ , projected-area disk average, then the instrument convolution and the planet-rotation convolution with the depth-dependent effective radius $R_{\text{eff}} = R_p \sqrt{(h + \delta)/\delta}$ (capped at the impact-parameter boundary), $\delta = (R_p/R_*)^2$. The three doublets are written under the same tpm_<line> naming as the H/He lines (§4.1), with the keys MgII, CaII, NaI. Alongside the spectra, the script prints the validated *single-component* depth table (line-center and 4 Å-band percentages for Mg II 2796, Ca II K, Na I D2), which is the quantity used for the Huang et al. (2023) comparisons; the doublet peak differs from the single-component value only by the partner-line wing (e.g. 3.75 vs. 3.72% for Mg II on WASP-121b). Two conventions matter when comparing with Huang: their table lists the effective transit radius

Table 1: Metal resonance doublets in EXHALE_transit.py (NIST).

species	component	λ_0 [Å]	f	A_{21} [s $^{-1}$]	window / R_{instr}
Mg II	k	2796.352	0.608	2.60×10^8	2790–2810 Å / 3×10^4
	h	2803.531	0.303	2.57×10^8	
Ca II	K	3933.663	0.6267	1.47×10^8	3927–3975 Å / Instr_res_Ha
	H	3968.469	0.3116	1.40×10^8	
Na I	D2	5889.951	0.641	6.16×10^7	5884–5902 Å / Instr_res_Ha
	D1	5895.924	0.320	6.14×10^7	

$R_{\text{eff}}/R_{\star} = \sqrt{(R_p/R_{\star})^2 + h}$, not a percent absorption, and Mg II is quoted in a 4 Å NUV bin while the optical lines are line-center (line-center vs. 4 Å differ by $\sim 3\times$ for the optically thick Mg II core).

For a Roche-lobe run the 3-D equipotential reconstruction (geometry = 'triaxial', roche_recon.py) replaces the spherical chords by the full Roche geometry with an LOS velocity for each sector $v_{\text{sub}}(r_{\text{eff}})x/r - \Omega y$ (wind + tidally locked rotation) and reports the same depth table.

7.2 Example spectra

Figure 1 shows the three doublet spectra for the WASP-121b worked example (backup/regression/tpm_wasp/, Roche Newton solution, metals on). The optically thick cores saturate, so the two doublet members reach similar depths despite the 2:1 oscillator-strength ratio; the rotation convolution mainly shallows and broadens the narrow Na I D cores.

8. Caveats

1. **$n = 2$ photoionization.** Γ_{2s}, Γ_{2p} are now estimated from a diluted stellar-blackbody Balmer continuum (§2.3, parameter T_star); this is a single blackbody approximation to the true stellar near-UV spectrum. Setting $T_{\text{star}} \leq 0$ reverts to the manual constants (default 0).
2. **Electron density.** $n_e = n_{\text{HII}} + n_{\text{HeII}} + 2n_{\text{HeIII}}$ from EXHALE is used (Christie assume $n_e = n_p$); the difference is small.
3. **$J_{\text{Ly}\alpha}$ convention.** The input must be the mean intensity J_{ν} at line center in cgs (§2.2). A different convention (energy density, frequency-integrated, etc.) requires rescaling the pump constant.
4. **Wavelength/damping.** $\lambda_{\text{H}\alpha}$ is the air value 6562.8 Å; the natural width is negligible relative to the thermal Doppler width, so the line is essentially Gaussian.

Reference

Christie, D., Arras, P., & Li, Z.-Y. 2013, ApJ, 772, 144: “H α Absorption in Transiting Exoplanet Atmospheres” (references/Christie_2013ApJ_772_144.pdf).

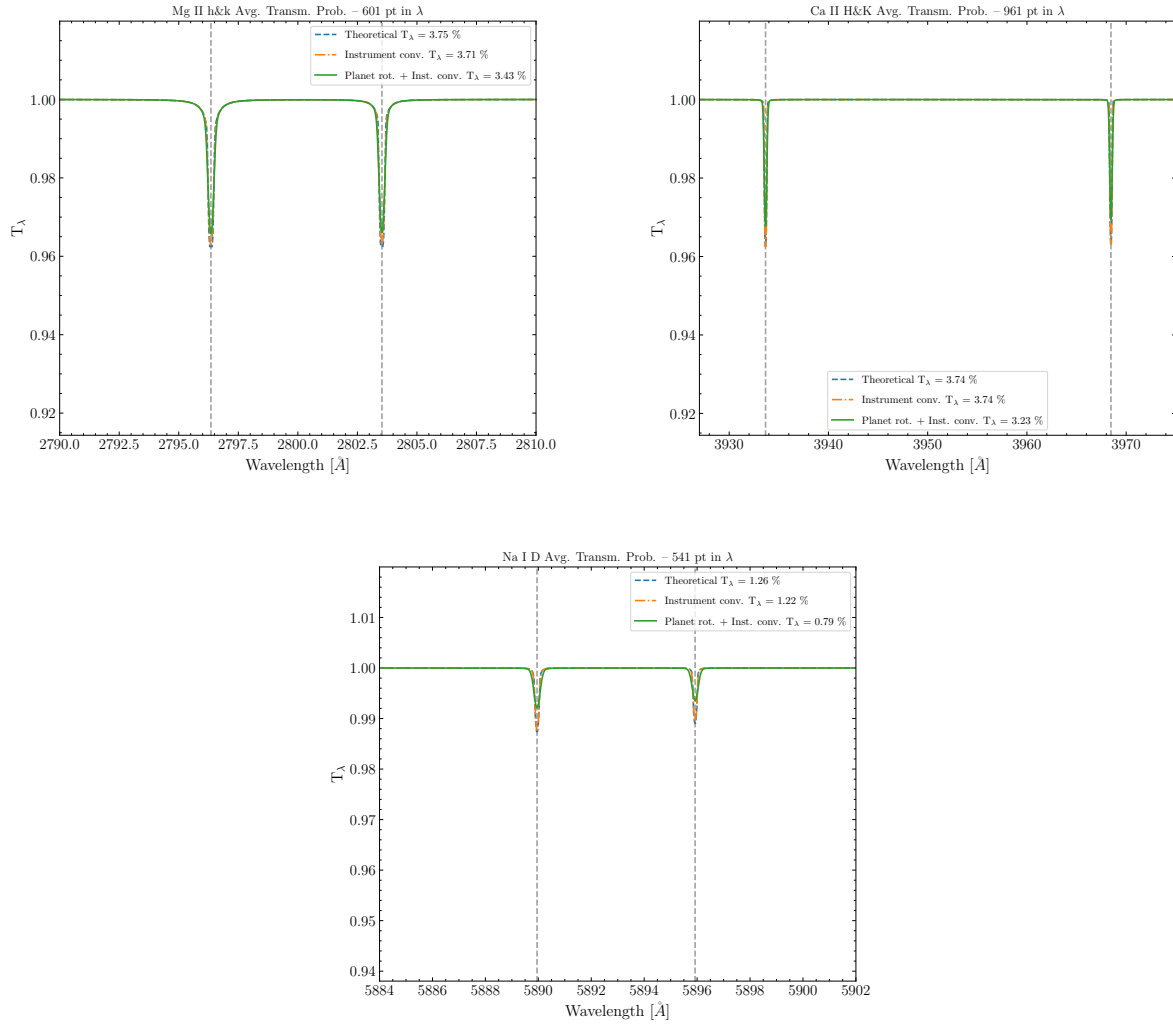


Figure 1: Metal resonance doublet transmission spectra for WASP-121b (backup/regression/tpm_wasp/): Mg II h&k (top left), Ca II H&K (top right), and Na I D (bottom). Each panel shows the theoretical, instrument-convolved, and instrument+rotation-convolved transmission, as for the H/He lines.

Huang, C., Arras, P., Christie, D., & Li, Z.-Y. 2017, ApJ, 851, 150: “A Model of the H α and Na Transmission Spectrum of HD 189733b” (references/Huang_2017_ApJ_851_150.pdf); the $J_{Ly\alpha}$ estimate is their Eq. (6) and related text.

Jensen, A. G., et al. 2012, ApJ, 751, 86: transit H α spectroscopy of HD 209458b and HD 189733b (comparison target for the HD 209458b notebook).